

Def. Let  $X$  be a Topological Space. A tuple of open sets  $\{U, V\}$  is called Separation of  $X$  if:  
 $U \neq \emptyset, V \neq \emptyset, U \cap V = \emptyset, U \cup V = X$ .

If the separation exists, then  $X$  is called disconnected.

Thm. If  $X$  is connected and  $f: X \rightarrow Y$  is continuous, then  $f[X]$  is connected subspace.

Proof. Suppose that  $f[X]$  is disconnected. Let  $\{U, V\} \subseteq \mathcal{O}_{f[X]}$  be a separation of  $f[X]$ .

That is,  $U \neq \emptyset, V \neq \emptyset, U \cap V = \emptyset, U \cup V = f[X]$ .

Then,  $f^{-1}[U] \neq \emptyset, f^{-1}[V] \neq \emptyset, f^{-1}[U \cap V] = f^{-1}[U] \cap f^{-1}[V] = \emptyset, f^{-1}[U \cup V] = f^{-1}[U] \cup f^{-1}[V] = X$ .

Since  $f$  is continuous,  $\{f^{-1}[U], f^{-1}[V]\}$  is a separation for  $X$ . Contradiction.

Thm. If  $A \subseteq X$  is connected and  $A \subseteq B \subseteq \overline{A}$ , then  $B$  is connected.

Proof. Suppose that  $B$  is disconnected. Then, there are opens  $U, V \subseteq X$  such that

$$U \cap B \neq \emptyset, V \cap B \neq \emptyset, U \cap V \cap B = \emptyset, U \cup V \supseteq B.$$

And, since  $B \subseteq \overline{A}$ ,  $U \cap \overline{A} \neq \emptyset$  and  $V \cap \overline{A} \neq \emptyset$ .

If  $U \cap A = \emptyset$ , then  $A \subseteq U^c$ , so  $\overline{A} \subseteq U^c$ . But this implies  $\overline{A} \cap U^c = \emptyset$ , contradiction.

Thus  $U \cap A \neq \emptyset$ , and similarly  $V \cap A \neq \emptyset$ . Moreover,

$$\emptyset = U \cap V \cap B \supseteq U \cap V \cap A, \text{ and } U \cup V \supseteq B \supseteq A, \text{ so}$$

$\{U, V\}$  is also separation of  $A$ , but it contradicts.

Thm. If subspaces  $A_\alpha \subseteq X$  ( $\alpha \in I$ ) are connected and  $\bigcap_{\alpha \in I} A_\alpha \neq \emptyset$ , then  $\bigcup_{\alpha \in I} A_\alpha$  is connected.

Proof. Denote  $A = \bigcup_{\alpha \in I} A_\alpha$ . Suppose that  $A$  is disconnected by the separation  $\{U, V\}$ .

That is,  $U, V$  are open s.t.  $U \neq \emptyset, V \neq \emptyset, U \cap V = \emptyset, U \cup V = A$ .

Let  $\alpha \in \bigcap_{\alpha \in I} A_\alpha$  be given. Then, for all  $\alpha \in I$ ,  $\alpha \in A_\alpha$ , thus  $\alpha \in A$ .

If  $\alpha \in U$ , put for each  $\alpha \in I$ ,  $U_\alpha = U \cap A_\alpha$  and  $V_\alpha = V \cap A_\alpha$ . Then,

$$\alpha \in U \cap A_\alpha \neq \emptyset, U_\alpha \cap V_\alpha = U \cap V \cap A_\alpha = \emptyset, U_\alpha \cup V_\alpha = (U \cup V) \cap A_\alpha = A_\alpha$$

Since  $A_\alpha$  is connected,  $V_\alpha = V \cap A_\alpha$  must be empty. I.e.,

$$V = V \cap A = V \cap \left( \bigcup_{\alpha \in I} A_\alpha \right) = \bigcup_{\alpha \in I} (V \cap A_\alpha) = \bigcup_{\alpha \in I} V_\alpha = \emptyset.$$

Contradiction. If  $\alpha \in V$ , it can be proved similarly.  $\square$

Corollary. If  $A_n \subseteq X$  ( $n \in \mathbb{N}$ ) are connected, and for all  $n \in \mathbb{N}$ ,

$$\left( \bigcup_{i=1}^n A_i \right) \cap A_{n+1} \neq \emptyset$$

then  $\bigcup_{n=1}^{\infty} A_n$  is connected.

Proof. Take  $B_n = A_1 \cup \dots \cup A_n$ . And using induction. (Use  $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$ )

Thm. Product of Connected space is Connected.

Proof. It holds in arbitrary product, but in this, prove only finite case  
Let  $X, Y$  be Connected Space. For any  $x \in X$  and  $y \in Y$ ,

$$X \times \{y\} \cong X \quad \text{and} \quad \{x\} \times Y \cong Y$$

are Connected. Moreover,  $X \times \{y\} \cap \{x\} \times Y = \{(x, y)\} \neq \emptyset$ . Therefore

$$(X \times \{y\}) \cup (\{x\} \times Y)$$

is Connected. Let  $b \in Y$  be fixed. Then, for any  $x \in X$ ,

$$T_x = (X \times \{b\}) \cup (\{x\} \times Y)$$

is Connected, and  $T_x \supseteq X \times \{b\}$ . Thus,  $X \times \{b\} \subseteq \bigcap_{x \in X} T_x \neq \emptyset$ .

Now,  $X \times Y = \bigcup_{x \in X} T_x$  is Connected.

4 Arbitrary Product Case

Let  $X_\alpha$  ( $\alpha \in I$ ) be Connected Spaces, and put  $X = \prod_{\alpha \in I} X_\alpha$ .

Let  $(b_\alpha)_{\alpha \in I} \in X$  be fixed. Then, for any finite subset  $\{\alpha_1, \dots, \alpha_n\} \subseteq I$ ,

$$X_{(\alpha_1, \dots, \alpha_n)} = \{(x_\alpha)_{\alpha \in I} \in X \mid \alpha \notin \{\alpha_1, \dots, \alpha_n\} \Rightarrow x_\alpha = b_\alpha\}.$$

Then,  $X_{(\alpha_1, \dots, \alpha_n)} \cong X_{\alpha_1} \times \dots \times X_{\alpha_n}$ , so it is Connected.

Moreover, for a Collection  $\mathcal{F}$  of every finite subsets of  $I$ ,

$\bigcap_{\{\alpha_1, \dots, \alpha_n\} \in \mathcal{F}} X_{(\alpha_1, \dots, \alpha_n)}$  contains  $(b_\alpha)$ , so non-empty.

$Y = \bigcup_{\{\alpha_1, \dots, \alpha_n\} \in \mathcal{F}} X_{(\alpha_1, \dots, \alpha_n)}$  is Connected.

So,  $Y$  is also Connected, enough to show  $X = Y$ .

Let  $\prod_{\alpha \in I} U_\alpha \subseteq X$  be a basis element of  $X$ . By def, all but finitely many  $U_\alpha = X_\alpha$ .

So, for some finite  $\{\alpha_1, \dots, \alpha_n\} \subseteq I$  s.t.  $U_\alpha = X_\alpha$  if  $\alpha \notin \{\alpha_1, \dots, \alpha_n\}$

then  $\prod_{\alpha \in I} U_\alpha \cap Y \neq \emptyset$ . So proved

Def. Let  $X$  be a Topological space. Define a relation  $\sim$  on  $X$  as:

$$x \sim y \Leftrightarrow \exists \text{ connected subspace } A \subseteq X \text{ s.t. } x \in A \text{ and } y \in A.$$

Then, it is equivalent relation; reflexivity and Symmetry are clear, and transitivity:  $x \sim y$  by  $A$  and  $y \sim z$  by  $B$ , then  $x \sim z$  by  $A \cup B$ .

Define Connected Component of  $X$  as:

$$C_x = [x]_{\sim}$$

Lemma. Let  $X$  be a Tps. Let  $x \in X$  be given.

1).  $C_x$  is the largest Connected subset containing  $x$ .

2). If  $A \neq \emptyset$  is connected subset of  $X$ , then there is Unique Connected Component  $C \supseteq A$ .

3). Connected Component is closed.

Proof of 1). Let  $A \subseteq X$  be a Connected subset containing  $x$ . Then,

$$y \in A \Rightarrow x \sim y \Rightarrow y \in C_x, \text{ so } A \subseteq C_x.$$

Moreover,  $C_x$  is Connected, because: given  $z \in C_x$ , there is a Connected subset  $A_z$  s.t.  $x, z \in A_z \subseteq C_x$ . Now, since  $\bigcap_{z \in C_x} A_z$  contains  $x$ , so nonempty, thus

$$C_x = \bigcup_{z \in C_x} A_z \text{ is Connected.}$$

2) is trivial. Fixed  $x \in A$ ,  $a \in A \Rightarrow a \in C_x$ . The uniqueness is given by the reflexivity

3).  $C_x \subseteq \overline{C_x}$  trivially, and  $\overline{C_x}$  is connected, so  $\overline{C_x} \subseteq C_x$ .

**Thm.** The Homeomorphism Preserves the Cardinality of Connected Components.

**Proof.** Let  $X$  and  $Y$  be Homeomorphic Topological Space with Homeomorphism  $f: X \rightarrow Y$ . Given  $x \in X$ , there is a unique Connected Component of  $x$ ,  $C_x \subseteq X$ .

Since the Connectedness is preserved by Continuous map,  $f[C_x]$  is Connected subset, containing  $f(x)$ . So,  $f[C_x] \subseteq C_{f(x)}$ . Similarly,  $f^{-1}[C_{f(x)}] \subseteq C_{f^{-1}(f(x))} = C_x$ .

Therefore, for any  $x \in X$ ,  $f[C_x] = C_{f(x)}$ . Similarly, given  $y \in Y$ ,  $f^{-1}[C_y] = C_{f^{-1}(y)}$ .

Now, define a map  $\phi: \{C_x | x \in X\} \rightarrow \{C_y | y \in Y\}; C_x \mapsto C_{f(x)}$

Then,  $\phi$  is one-to-one:

$$C_{f(x_1)} = C_{f(x_2)} \Leftrightarrow f[C_{x_1}] = f[C_{x_2}], \text{ since } f \text{ is Homeo, } C_{x_1} = C_{x_2}$$

(Take  $f^{-1}$  to the both sides.) And, given  $C_y$  from the Co domain, there is  $x \in X$  st  $y = f(x)$ . So  $\phi(C_x) = C_{f(x)} = C_y$ .  $\square$

**Thm.** If a Topological Space  $X$  has only finite Connected Components, then every Components are Clopen.

**Proof.** Let  $C_1, \dots, C_n$  be all Connected Components of  $X$ . Then for each  $i$ ,

$$C_i = X \setminus \left[ \bigcup_{j \neq i} C_j \right]$$

is open.

Def. Let  $X$  be a Topological space.

We said  $X$  is locally Connected at  $x \in X$  if:

For any open  $N \subseteq X$  containing  $x \in X$ , there is connected open  $V$  s.t.  $x \in V \subseteq N$ .

$X$  is called locally Connected, if it is locally connected at every point.

Thm.  $X$  is locally Connected if and only if:

for any open  $U \subseteq X$ , every Connected component of  $U$  is open.

Proof. Suppose that  $X$  is locally Connected space. Given an open set  $U \subseteq X$ , let  $C \subseteq U$  be a Connected component of the subspace  $U$ . Since  $X$  is l.c. connected, given  $x \in C$ , there is a Connected open set  $V_x$  s.t.  $x \in V_x \subseteq U$ .

Since  $C$  is Connected component of  $x$ ,  $V_x \subseteq C$ . So,

$$C = \bigcup_{x \in C} V_x \text{ is open set.}$$

Conversely, let  $x \in N$  be given. Then, there is open Connected component  $C_x \subseteq U$  exists.

Above definition and theorem are same in the case of Path-Connected.

Thm. If  $X$  is locally path-connected, then the Connected component and path component are same.

Proof. Let  $C \subseteq X$  be a Connected component. Given  $x \in C$ , let  $P_x$  be a path component of  $x$ .

Since  $P_x$  is Path-Connected, so Connected, thus  $P_x \subseteq C$ .

To show  $P_x = C$ , Suppose that  $P_x \subsetneq C$ . Let  $\mathcal{Q}$  be the union of:

$$\{P_y \mid P_y \text{ is a path-component of } X \text{ s.t. } P_y \not\subseteq P_x \text{ and } P_y \cap C \neq \emptyset\}.$$

Then, for each  $P_y$  is contained in  $C$ , so that  $C = P_x \cup \mathcal{Q}$ .

Since  $X$  is locally path connected,  $P_x$  and  $\mathcal{Q}$  are open, so these become the separation for  $C$ , it is contradiction.

Corollary. If  $X$  is locally path-connected, then

Path Connected  $\Rightarrow$  Connected.

Thm. Quotient map preserves locally connected by image.

Proof. Let  $X$  be a locally connected and  $p: X \rightarrow Y$  be a quotient map.

Let  $C \in Y$  be a connected component of an open set  $U$  of  $Y$ .

Then, clearly  $p^{-1}[C] \subseteq p^{-1}[U]$ . Given  $x \in p^{-1}[C]$ , there is a connected component  $C_x \ni x$ .

Then,  $p[C_x] \in p[p^{-1}[C]]$  is connected, so  $p[C_x] \subseteq C$ , or  $C_x \subseteq p^{-1}[C]$ .

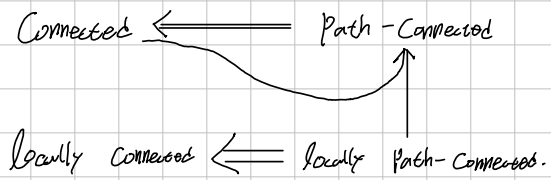
So,  $p^{-1}[C] = \bigcup_{x \in p^{-1}[C]} C_x$  and since  $C_x$  are components of open set  $p^{-1}[U]$ ,

So these are open, since  $X$  is locally connected.

Now,  $p^{-1}[C]$  is open, so is  $C$  open, since  $p$  is quotient map.  $\square$

Thm. Quotient map preserves locally path-connected by image.

Summary,



Def. Let  $X$  be a Topological space.

If for any  $a, b \in X$ , there is a continuous map

$$p_a^b: [0, 1] \rightarrow X \text{ s.t. } p_a^b(0) = a \text{ and } p_a^b(1) = b$$

then  $X$  is called path-connected.

Thm Path-Connected  $\Rightarrow$  Connected.

Proof. Suppose that  $X$  is disconnected, but path-connected.

Let  $\{U, V\}$  be the separation for  $X$ . Given  $a \in U$  and  $b \in V$ ,

there exists a path  $p: [0, 1] \rightarrow X$  s.t.  $p(0) = a$ ,  $p(1) = b$ .

Since  $p$  is continuous and  $[0, 1]$  is connected,  $\text{Im } p$  is connected, but

$$U \cap \text{Im } p \ni a, \quad V \cap \text{Im } p \ni b, \quad U \cap V \cap \text{Im } p = \emptyset, \quad U \cup V \supseteq \text{Im } p$$

so,  $\{U, V\}$  is the separation for  $\text{Im } p$ , it is contradiction

Thm. Let  $X$  be a space. If  $A_\alpha \subseteq X$  ( $\alpha \in \mathcal{A}$ ) are path-connected with  $\bigcap_{\alpha \in \mathcal{A}} A_\alpha \neq \emptyset$ , then  $\bigcup_{\alpha \in \mathcal{A}} A_\alpha$  is path-connected.

Proof.

## Topologist's Sine Curve.

Def. In  $\mathbb{R}^2$ , let  $L = \{0\} \times [0, 1]$  and  $S = \{(x, y) \mid x > 0, y = \sin \frac{1}{x}\}$ .

Then, the subspace  $L \cup S$  is called the Topologist's Sine Curve.

It has the following properties:

1). Connected.

Since  $x \mapsto \sin \frac{1}{x}$  ( $x > 0$ ) is continuous, the graph  $S$  is connected.

And, since  $L \cup S \subseteq \overline{S}$ , ( $L \subseteq \overline{S}$ , so  $L \cup S \subseteq \overline{S}$ ),

$L \cup S$  is also connected, by  $A$  connected  $\Rightarrow \overline{A}$  is connected.

2. Not path-connected.

$S$  is path-connected, since  $(0, \infty)$  is path-connected and  $x \mapsto \sin \frac{1}{x}$  is continuous.

But, for  $(0, 0) \in L$  and  $(1, \sin 1) \in S$ , there is no path between  $(0, 0)$  and  $(1, \sin 1)$ .

(This is also the counter-example for  $S$  path. conn. but  $\overline{S}$  is not.)

suppose that there is a path  $p: [0, 1] \rightarrow L \cup S$  from  $(0, 0)$  to  $(1, \sin 1)$ .

Let  $t_0 = \sup \{t \in [0, 1] \mid p(t) \in L\} = \sup p^{-1}[L]$ .

Since  $L$  is compact,  $t_0 \neq 1$ . (If  $t_0 = 1$ , then  $p(t)$  cannot be  $(1, \sin 1)$ ).

Meanwhile, for any  $\delta > 0$ , there is  $t_0 < t' < t_0 + \delta$  such that

$$|\pi_y(p(t_0)) - \pi_y(p(t'))| \geq \frac{1}{2}, \text{ so}$$

it is contradiction to  $p$  is continuous. ( $\pi_y$  is a projection onto  $y$ )

